Little Z's Function

Input file:	`standard input`
Output file:	`standard output`
Time limit:	5.5 seconds
Memory limit:	1024 megabytes

Little Z was obsessed with number-theoretic functions such as Euler's totient function φ* back in high school.

One day, Little Z decides to play a game with his robotic friend. First, Little Z gives a permutation† p of length n. Then he asks his friend multiple queries. For a given interval $[l, r]$ $(1 \le l < r \le n)$, find the maximum value of $\varphi(p_i \times p_j)$ where $l \le i < j \le r$.

Formally, for each query with two integers l, r, you need to compute:

$$\max_{l \le i < j \le r} \varphi(p_i \times p_j)$$

Little Z has solved exactly 0 queries. Please help him compute the remaining q queries.

Input

The first line contains a positive integer n $(2 \le n \le 5 \times 10^5)$ — the length of the permutation.

The second line contains n distinct integers $p_1, p_2, \ldots, p_n$ $(1 \le p_i \le n)$ — the given permutation p.

The third line contains a positive integer q $(1 \le q \le 5 \times 10^5)$ — the number of queries.

Each of the next q lines contains two integers l_i and r_i $(1 \le l_i < r_i \le n)$ — the endpoints of the i-th query.

Output

For each query, output a single integer — the maximum value.

Example

standard input	standard output
6 3 2 5 4 1 6 8 4 5 2 6 3 5 2 3 3 4 1 5 4 5 2 6	2 8 8 4 8 8 2 8

*$\varphi(n)$ denotes the number of positive integers not exceeding n that are coprime to n.

†A permutation of length n is an array consisting of n distinct integers from 1 to n in arbitrary order. For example, $[2, 3, 1, 5, 4]$ is a permutation, but $[1, 2, 2]$ is not a permutation (2 appears twice in the array), and $[1, 3, 4]$ is also not a permutation ($n = 3$ but there is 4 in the array).